線上聊天室

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1. 技術與框架
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目錄

技術與框架

技術

- Node.js
- 一種是能夠在伺服器端運行 JavaScript 的開放原始碼、跨平台 JavaScript 執行環境。
- ngrok

框架

- Socket.io：一个面向实时 web 应用的 JavaScript 库
- Express：用於client跟server獲取資料

相關程式碼-Server

- 設立一個變數express來導入Express模塊，並用變數app建立一個Express應用
- 設立一個變數serv來讓app可以執行HTTP操作
- 變數io為導入socket.io框架，並綁定到服務器上
- app.get用來指定路徑，回傳HTTP 的GET請求
- serv.listen與3000端口上啟動服務器

a.js

```
var express = require('express');
var app = express();
var serv = require('http').Server(app);//Expres
var io = require('socket.io')(serv);

app.get('/', function(req,res){//client ask ser
  res.sendFile(__dirname + '/index.html');//ser
});
app.use('/', express.static(__dirname + '/'));

serv.listen(3000);//make http server listen on
console.log("Server started");//server reply
```

相關程式碼-Server

- socket.emit根據事件名，觸發相關function， 並將資料回傳
- socket.on監聽emit發射的事件， 並執行相關function

```
25  io.on('connection', function(socket){//listen on the connection
26    console.log('new user connected');
27
28    var user = null;
29
30    socket.on('new-user', function(data){
31      user = data.username;
32      if(checklist(user)){//if name already used return false
33        socket.emit('check-list', {success:false});
34      }
35      else{// add new user into array
36        socket.emit('check-list', {success:true});
37        users.push(user);
38        console.log(user + ' connected');
39        console.log(users);
40        socket.broadcast.emit('new-user', user + ' connected');
41        updateList();
42      }
43    });
44
45    socket.on('disconnect', function(){// show who left chat
46      console.log(user + ' disconnected');
47      socket.broadcast.emit('user-left', user + ' disconnected');
48      console.log(users);
49      users.splice(users.indexOf(user),1);
50      updateList();
51    });
52
53    socket.on('new-msg', function(msg){
54      io.emit('new-msg', '<b>' + user + '</b>'  + ': ' + msg);//se
55    });
56
57    function updateList(){//sent to client to display who's online
58      io.emit('update-list', users);
59    }
60
61  });
```

相關程式碼-Server

- 設立一個陣列users來讀取使用者名字
- 利用checklist來製作一個檢查陣列的function

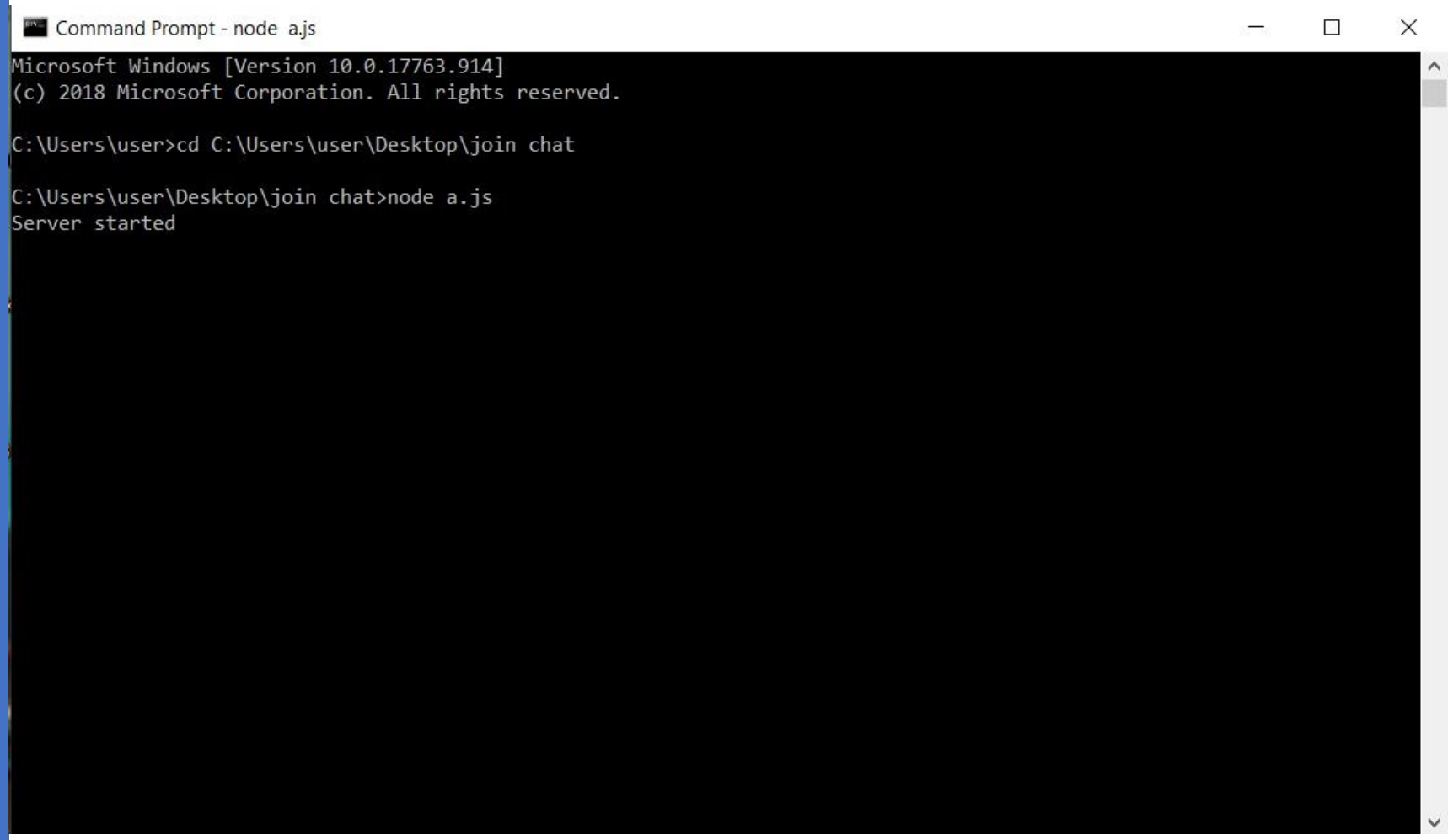
```
var users = [];//users in server

var checklist = function(data){//check if username is already used
  for(var i = 0;i<users.length;i++){
    if(users[i] == data){
      return true;
    }
  }
  return false;
}
```

相關程式碼-Client

- 設立一個變數socket來與服務端建立連接
- 通过引入 <script src="/socket.io/socket.io.js"></script> 即可使用socket.io
- 使用者可以執行emit與on的操作

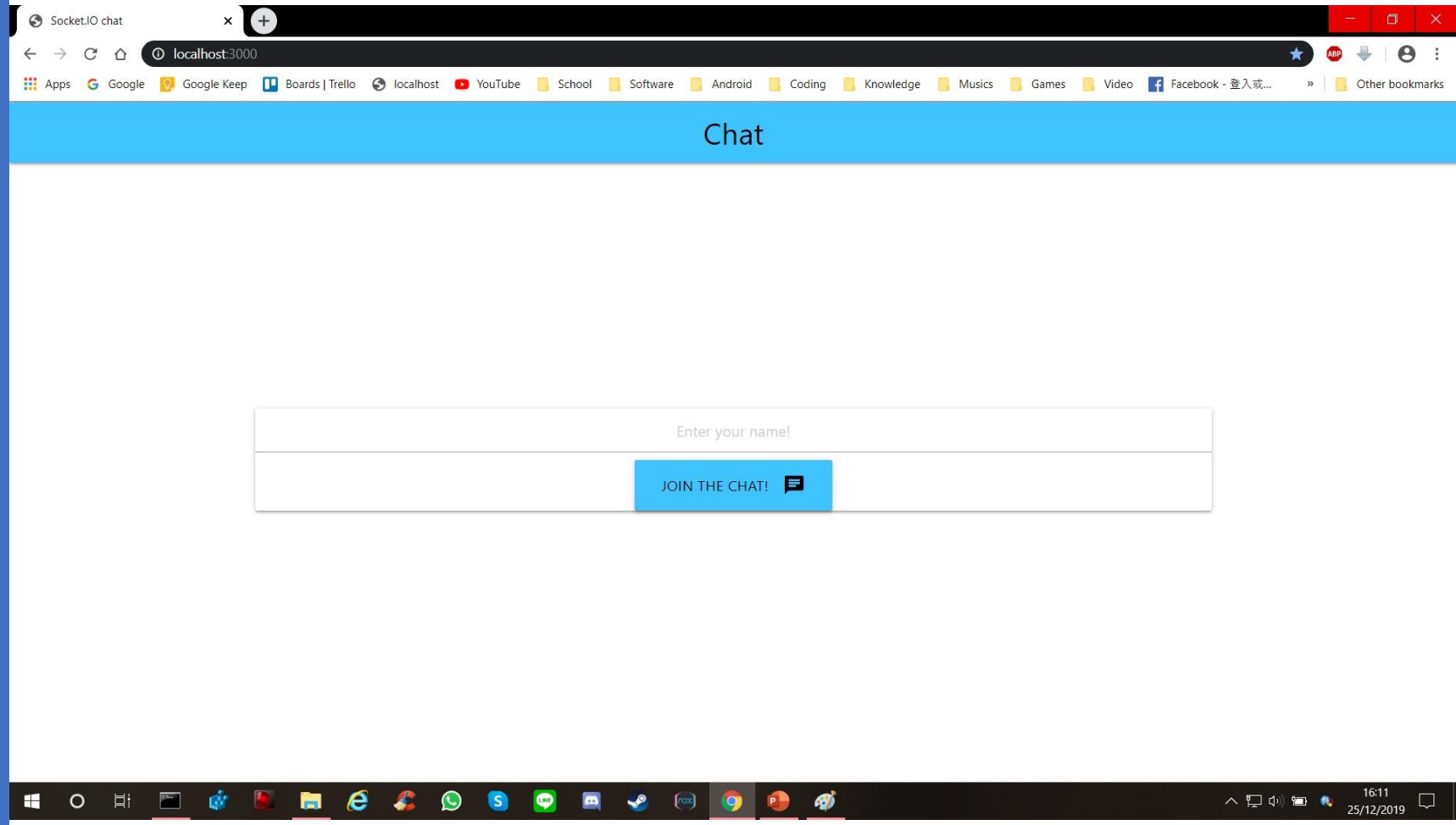
```
9   <script src="/socket.io/socket.io.js"></script>
10
11  <script>
12  window.onload = function(){
13    var socket = io();// connect to server on port 3000
```

執行畫面



- 打開terminal移動到相關路徑輸入node a.js
- 成功後會回傳Server started

執行畫面



- 利用遊覽器移動到http://localhost:3000/就會看到網頁頁面

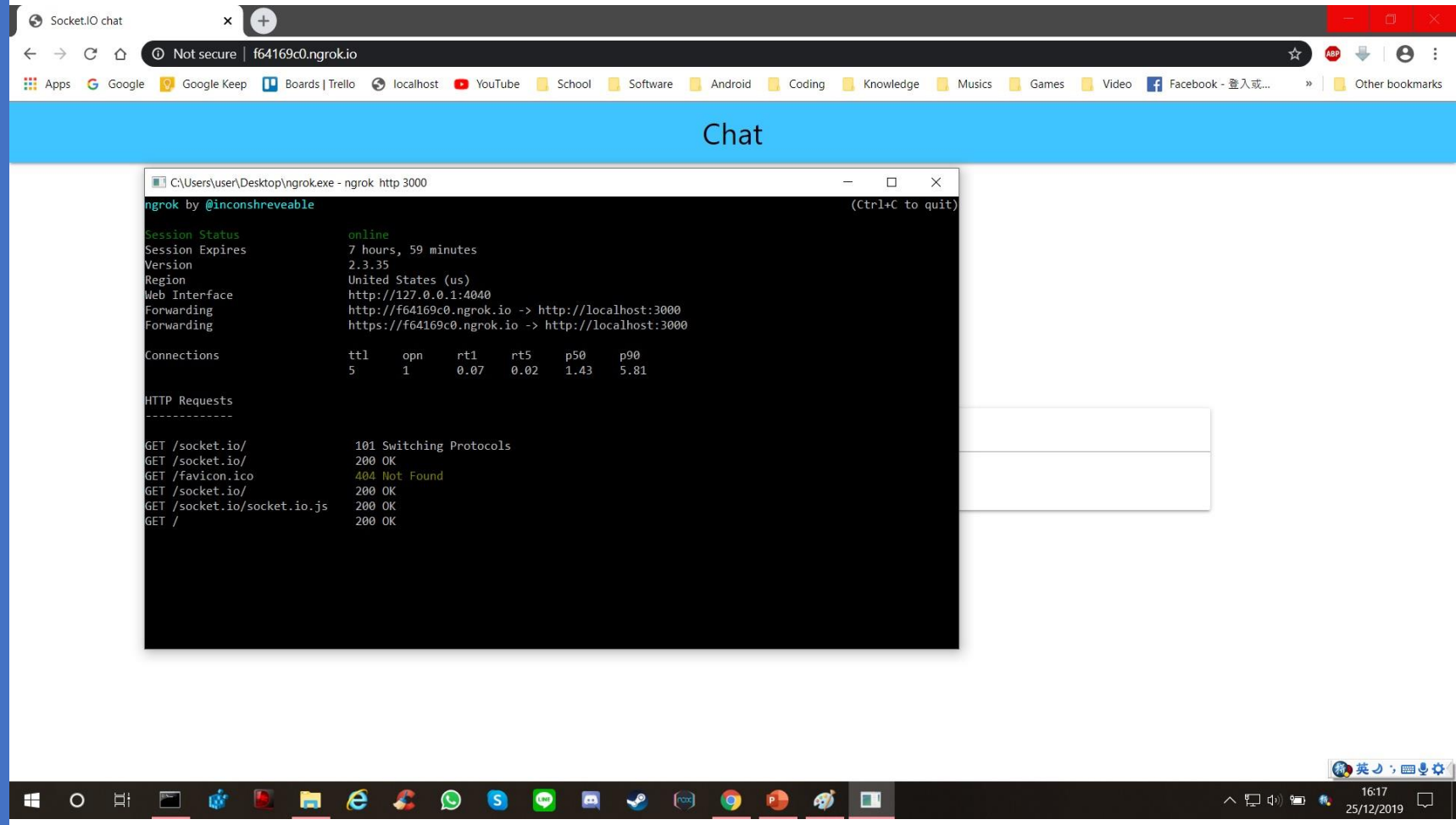
執行畫面

C:\Users\user\Desktop\ngrok.exe

```
EXAMPLES:
    ngrok http 80                    # secure public URL for port 80 web server
    ngrok http -subdomain=baz 8080   # port 8080 available at baz.ngrok.io
    ngrok http foo.dev:80            # tunnel to host:port instead of localhost
    ngrok http https://localhost     # expose a local https server
    ngrok tcp 22                     # tunnel arbitrary TCP traffic to port 22
    ngrok tls -hostname=foo.com 443  # TLS traffic for foo.com to port 443
    ngrok start foo bar baz          # start tunnels from the configuration file

VERSION:
   2.3.35

AUTHOR:
  inconshreveable - <alan@ngrok.com>

COMMANDS:
   authtoken    save authtoken to configuration file
   credits      prints author and licensing information
   http         start an HTTP tunnel
   start        start tunnels by name from the configuration file
   tcp          start a TCP tunnel
   tls          start a TLS tunnel
   update       update ngrok to the latest version
   version      print the version string
   help         Shows a list of commands or help for one command

ngrok is a command line application, try typing 'ngrok.exe http 80'
at this terminal prompt to expose port 80.
C:\Users\user\Desktop>ngrok http 3000
```

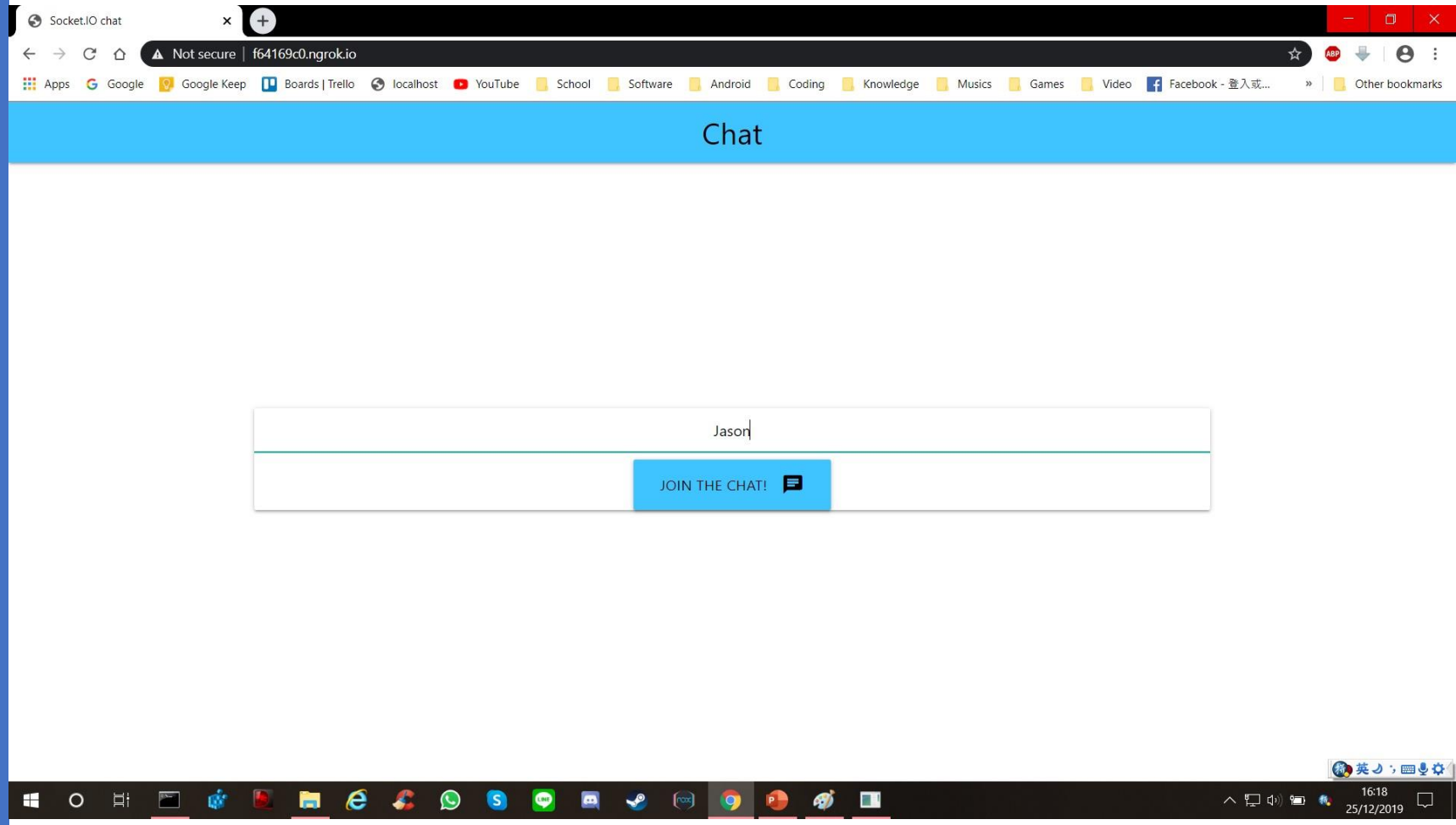
- 打開ngrok.exe， 並輸入ngrok http 3000
- 使用ngrok指定到電腦的port 3000， 可以把外界的請求轉發到你指定的 Port

執行畫面



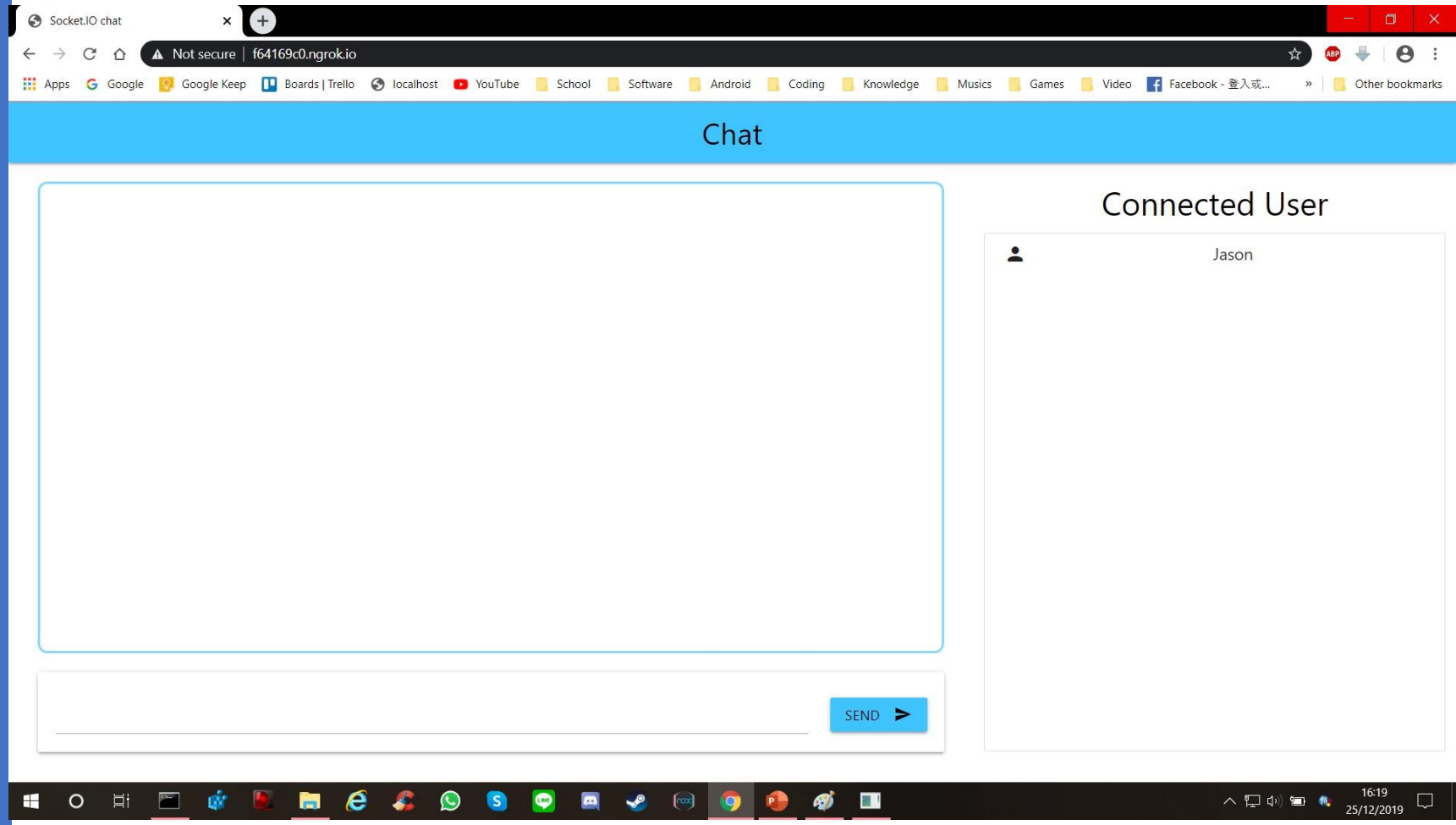
- ngrok會回傳一個網址，打開來就是網頁聊天頁面

執行畫面



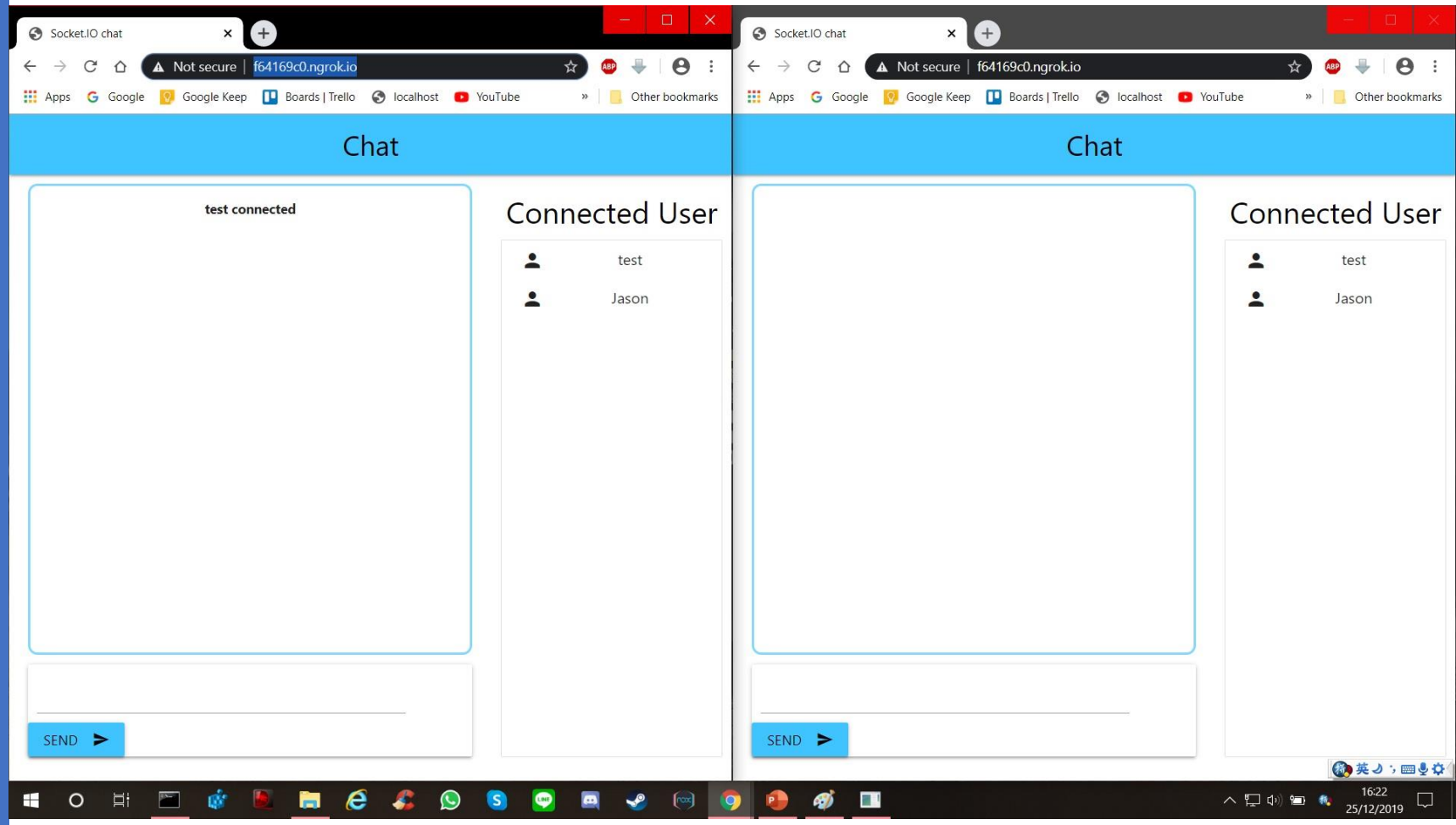
- 此時可以使用頁面輸入名字並進入到聊天室

執行畫面



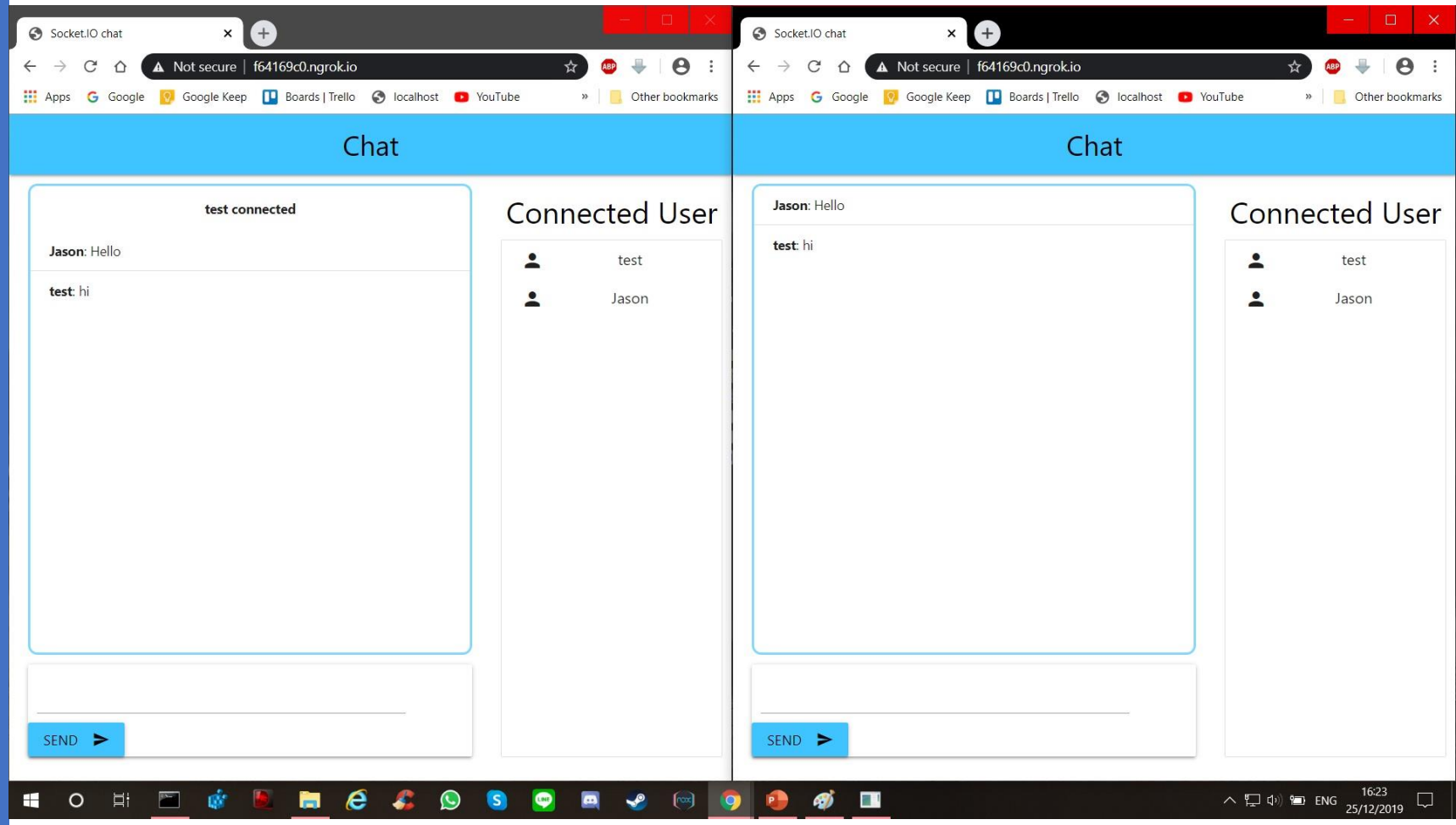
- 聊天室的左方會顯示在線使用者

執行畫面



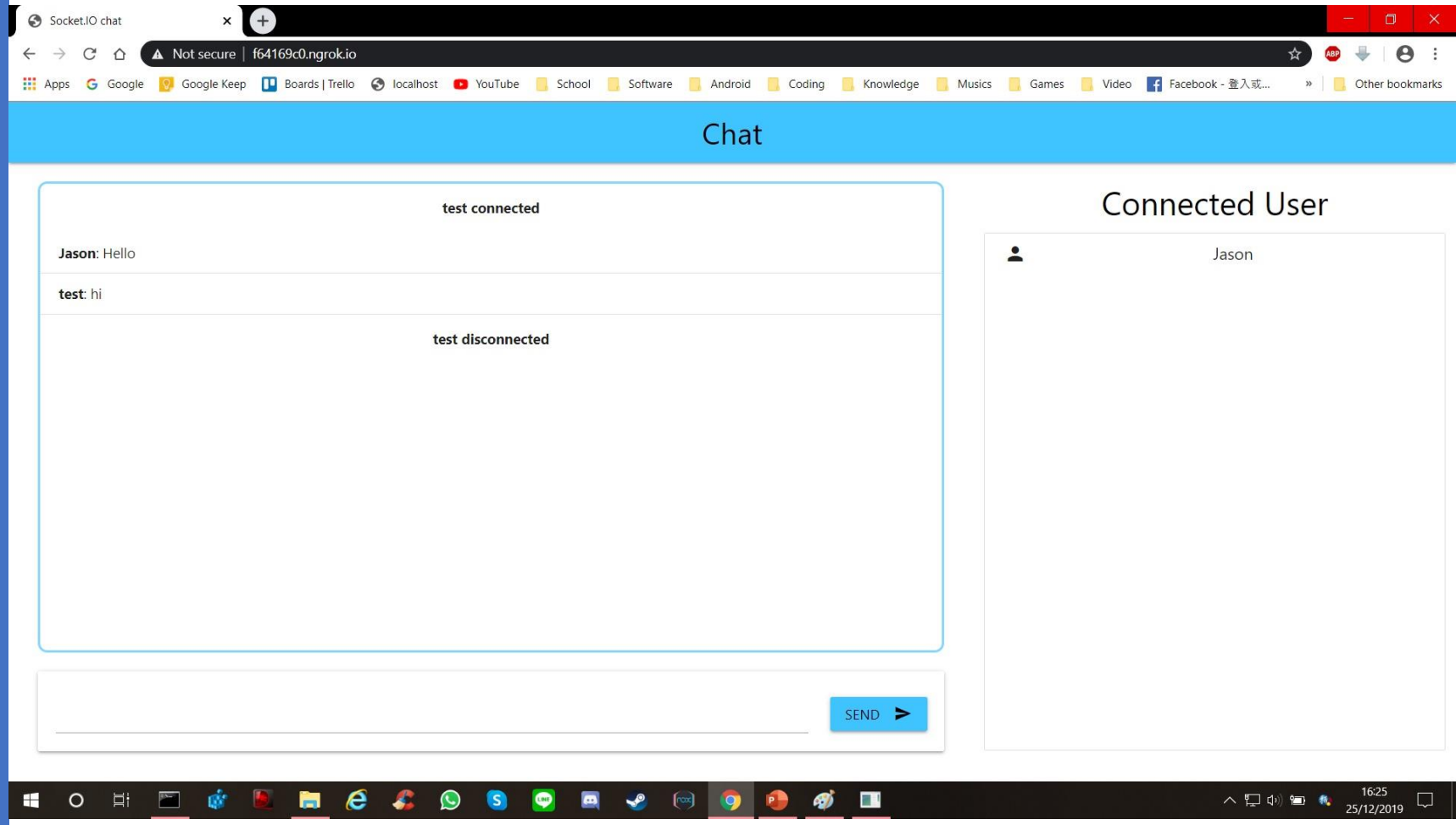
- 當另一個使用者加入時，會顯示新使用者加入，左邊的在線用戶也會更新

執行畫面



- 輸入的訊息都會同步顯示在頁面上

執行畫面



- 當另一個使用者離開時，會顯示使用者離開，並更新左邊的在線用戶

執行畫面

Command Prompt - node a.js

```
Microsoft Windows [Version 10.0.17763.914]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\user>cd C:\Users\user\Desktop\join chat

C:\Users\user\Desktop\join chat>node a.js
Server started
new user connected
null disconnected
[]
new user connected
Jason connected
[ 'Jason' ]
new user connected
test connected
[ 'Jason', 'test' ]
test disconnected
[ 'Jason', 'test' ]
Jason disconnected
[ 'Jason' ]
```

- 在terminal裡會顯示使用者加入與離開的訊息

THE END